Forecasting

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Maybe I should do this in RMarkdown instead...

1 Statistics stuff

- Forecasting means prediction
- Predicting a single number tells you nothing because we are dealing with probabilities and randomness, so always produce a confidence interval.
- Pick a model, then residual diagnostics, if they pass you can do inference (prediction, condifence intervals, p-values).
- The idea behind residual diagnostics is to see if the random component of your model is truly random and does not include deterministic patterns you should be using in the model.
- $Cov(X_i, X_j) = 0$ for uncorrelated RVs
- for two RVs, Independence is a stronger condition than non-correlation
- WHAT ARE DEGREES OF FREEDOM?

2 Introduction to Forecasting

Definition 1. *Time series: A sequence of random variables indexed by time* $\{X_t : t \in T \subset \mathbb{N}\}$*. It is a special case of a stochastic process.*

Definition 2. *Stochastic process: a collection of random variables (they don't need to have the same distribution!) indexed by a set* I*.* I *could be continuous or discrete.*

We often want to model time series data with *non-stationary trends*:

- Seasonality
- trending mean
- Non-constant variance (fanning in our out pattern)

Ideally given a time series $\{X_1, X_2, ...X_n\}$ we would know the *joint distribution* of such vectors.

$$F(x_1, x_2, \dots x_n) = P(X_1, X_2, \dots X_n)$$

This is because individual (marginal) distributions of X_i don't tell you about the combined behaviour.
We want to study a stochastic process as a mathematical which is a sequence of RV's with some predictable distribution ie not just a vector of n random and unrelated numbers. As such, the existence of a stochastic process should always be questioned. Kolmogorov's theorem tells us about the existence of discrete time stochastic process build from $X_i's$ as long as the X_i satisfy the two xonsistency conditions.
If all finite dimensional distributions exist, and they are consistent with each other under marginalization, then the stochastic process exists. Then **we can use finite dimensional models ot define a general process**.*****
All finite dimensional distributions of $X_1, X_2, \dots X_n$ are those of all non-empty subsets of the sequence ie $\{X_1, X_2, \dots X_n\}^*$ so it is impossible to have all of these in practice since, if we are sampling, we don't even have the distribution of a single X_i and must do with estimates.
Fortunately, in most application we can get by with knowing $E[X_t], E[X_t^2], E[X_t X_j]$. Which is equivalent to knowing $E[X_t], Var(X_t), Cov(X_t, X_j)$ for **ALL** indices.

Definition 3. ***A time series model*** *for the observed data $\{x_t\}$ is a specification of the joint distributions or more often in practice, the means and covariances of a sequence of random variables $\{X_t\}$ of which $\{x_t\}$ is a realization.*

2.1 Zero-mean time series models

2.1.1 i.i.d Noise

Definition 4. ***iid Noise*** *a sequence $\{X_i : i = 1, 2 \dots n\}$ with:*

- *independence property is satisfied: joint = product of marginals.*
- $E[X_i] = 0 \forall i$

As is the case with a discrete time stochastic process with independent components, the memory-less property is satisfied.

Proof.

$$P(X_{n+h} \leq x | X_1 \leq x_1, \dots, X_1 \leq x_n) = \frac{P(X_{n+h} \leq x) F(x_1, \dots x_n)}{F(x_1, \dots x_n)}$$

□

Note that in the nominator above, the probability of the intersection of events is equal to the individual probabilities multiplied by independence.
Also note how we can have the individual X_i be continuously distributed but discrete stochastic process (indexed by the naturals)

2.1.2 White Noise

Sequence of uncorrelated random variables $\{X_1, X_2, \dots\}$ with zero mean and finite variance. Denoted $X_i \sim WN(0, \sigma)$

- $E(X_i) = 0$
- Var
- $Cov(X_i, X_j) = 0$ for uncorrelated

2.1.3 Random Walk

Definition 5. *A R.V which is a* ***series*** *of* ***iid noise OR White noise*** *distributed R.V.s* $S_t = \sum_{i=1}^{t} X_i$*. Usually starting at zero.*

What is the relationship between *white noise* and *iid noise*? Independence is a stronger condition than non-correlation. For independence we need to know joint distributions but for non-correlation we don't. So we'd rather work with white noise.

3 Classical decomposition

In regression we are always modelling $Y|X$ and not just Y since we are actually using the explanatory variable to fit Y.
The general framework is to remove the **trend**, **seasonality**, and **non-constant variance** patterns in the data so that *hopefully* we are left with a stationary process. Then model the detrendized and deseasonalized data.

3.1 Additive classical decomposition

Fitting models with trend and seasonality:

$$X_t = m_t + S_t + Y_t$$

where,

- X_t is value of the process at time t
- S_t models the periodic deterministic component of X_t at time t
- m_t is the TREND deterministic component. Doesn't have to be linear.
- Y_t is the stochastic component, white noise

Note that the expected value of the process depends on time: $E[X_t] = E[S_t] + E[m_t]$. Particularly if the expectation for the trend component is non-zero as the expectation of periodic functions is often constant.
We don't need Fourier series to model periodic patterns (in discrete stochastic processes), we can figure out the period d and then use $d-1$ indicator variables

whose coefficients give the period height at each point of time. Let's assume the period for some time series is 12 corresponding to 12 months, we use $d-1=11$ indicator variables $X_1, X_2, \ldots, X_11$ to represent:

$$x_1 = \{\begin{cases} 1, & \text{if month is February} \\ 0, & \text{else} \end{cases} \tag{1}$$

The reason we don't need as many **indicator variables** as the period is that we can measure the first point (January) with just trend $X_1 = m_t + Y_t$ and then have the remaining $d-1$ coefficients β_j tell us the deviation of point X_{i+1} relative to X_1 it is the baseline.
We use regression for m_t and S_t, nothing new so far. The trend m_t need not follow a linear regression model, we can model it to be any regular function ie. $m_t = \beta_0 + \exp \beta_1 t$.
If we have monthly data, the period need not be 12, it might be biweekly if we are measuring something that is correlated to paydays.

4 Loss Functions/Decision Theory

Measure the quality of the fit. As you imagine, there are several ways to measure "quality". We need a measure to compare the fitted predicted values to the truth.
Denote the loss function as $L(Y, f(X))$. It measures the error in estimating Y with the estimate $f(X)$
The loss function **is a random variable**. The expected value of the loss function with respect to the **joint probability function** is called the **risk**:

$$E_{X,Y}[L(Y, f(X))]$$

A couple of things here:
The expectation is over the joint distribution in theory, but in practice, since we are tryint to use X to predict Y we *assume X is given* so the expecation will be ralative to this distribution ie.

$$E_{Y|X}[L(Y, f(X))|X = x] = \int_{\mathbb{R}} P(Y|X = x) dL(Y, f(X))$$

Even these conditional probabilities are hard to wrap your mind around, we have to assume their distribution, often to uniform when we consider the **empirical** risk estimate:

$$\hat{E}[L(y_i, f(x_i))] = \sum_{\Omega} L(y_i, f(x_i))$$

4.0.1 Quadratic Loss/Squared error loss

$L_{SE}(Y, (f(X)) = (Y - f(X))^2$ where $f(X) = \hat{Y} :=$ the random variable Y estimated as a function of X.
The risk of this loss function also has a special name: **MSE**

$$MSE(f) = E_{X,Y}[L_{SE}(Y, f(X))]$$

Theorem 1. ***Conditional Expectation Theorem****: If X, Y are two random variables with $E(Y) = \mu, Var(Y) < \infty$, then the function f which minimizes $MSE(f)$ for X, Y is given by the conditional expectation:*

$$f(X) = E_{Y|X}(Y|X)$$

Proof. **********

- let $f(X) = c$ and show that the constant c that minimizes the risk is the mean of Y.
- Assume $c(x)$ minimizes

□

Notes:

- $E(Y|X = x)$ is a number depending on x. So it's a function of x
- $E(Y|X)$ is a random variable

4.0.2 Absolute Error Loss

Minimized by the median

5 Regression

The MSE is minimized by the conditional expection $E(Y|X)$. But we don't really know how to get this, so we take a step back and assume the form of $f(X) = \boldsymbol{X\beta} + \epsilon$ in the risk minimization problem (called OLS, RSSQ) and then solve for the coefficients using matrix methods.
In multiple regression we have some simplifying assumptions s.t.

$$f(X; \boldsymbol{\beta}) = E[Y|X] = \beta_0 + \sum_{i=0}^{n} \beta_i X_i = \boldsymbol{X\beta} + \epsilon$$

Where X is the $n \times (p-1)$ design matrix.

5.0.1 OLS

We call the risk under the assumed linear model the **Residual Sum of Squares**. To estimate the vector of parameters $\boldsymbol{\beta}$ using $\{(y_i, \boldsymbol{x}_i)|i \in [n]\}$ where $x_i = (x_{i1}, x_{i2}, \dots x_{ip})$ represents a row on the design matrix where each j column is a variate (rooms, bathrooms, parking spots) or an attribute (bathrooms squared x_i^p) of the $i-$th observation/unit.

$$RSS(\boldsymbol{\beta}) := \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij})$$

As a function of the coefficient vector, our estimate for $\hat{\beta}$ in quadratic matrix form is:

$$\hat{\beta} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}}(\boldsymbol{y} - \boldsymbol{X\beta})^T(\boldsymbol{y} - \boldsymbol{X\beta})$$

The OLS extimate for the coefficient vector is:

$$\boldsymbol{\beta}_{OLS} = (X^TX)^{-1}X^T$$

Sampling distribution The sampling distribution of the OLS coefficients estimator is $\tilde{\beta}_{OLS} \sim MVN(\beta, \sigma(X^TX)^{-1})$
The variance estimator also has a sampling distribution following a chi-squared. The test of hypothesis for $H_0 : \hat{\beta}_j = 0$ uses the pivota quantity

$$z_j = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p-1}$$

Where v_j is the $(j+1)^{th}$ diagonal element of $(X^TX)^{-1}$
Properties of the OLS estimator $\tilde{\beta}$

- it is unbiased ie. $E(\tilde{\beta}) = \boldsymbol{\beta}$
- variance covariance matrix is $Var((X^TX)^{-1}X^T) = \sigma^2(X^TX)^{-1}$. This means that the variability and thus length of confidence and prediction intervals depends on σ (which we don't control that much) and on the design matrix X which we can indeed control. The size of the variance will explode if $(X^TX)^{-1}$ explodes. This happens when columns of X are *correlated*, ie. you have *redundant* information in the design matrix.
- σ^2 is estimated with the unbiased estimator $\tilde{\sigma}^2 = \frac{SSE}{n-p-1}$ recall p is the number of exploratory variates.
- a $100(1-\alpha)\%$ *confidence interval* for β_i is $\hat{\beta}_i \pm t_{\alpha/2,n-p-1}SE(\hat{\beta}_i)$ where the Standard Error of the estimate $SE(\hat{\beta}_i)$ is the square root of the corresponding diagonal element of the variance covarinace matrix $\sigma^2(X^TX)^{-1}$.

You can also show OLS with maximum likelihood estimation.

Sum of squares decomposition We can show that TSS = SSReg + SSE
We consider the hat matrix $H = X(X^T X)^{-1} X^T$ which:

- is an *orthogonal projection* matrix since it is indempotent.
- is indempotent, so $H^2 = H$
- the fitted values are the projection of the observed values in the training set y onto the linear subspace spanned by the columnds of $\mathbf{X}$. ie $\hat{y} = Hy$
- the estimated residuals are the corresponding orthogonal complement $\hat{r} = y - Hy = (I - H)y$

5.0.2 Assumptions on the residuals

- The error terms and thus the Y_i are independent.
- The error tersm and thus Y_i are normally distributed (Not uniformly if you want to estimate conditional probabilities for some reason)
- the error terms and thus Y_i have a constant variance.
- the erro terms have a constant mean of 0. Not Y_i necessarily though.n

5.0.3 Regression in R

Consider the output of *lm*.

6 Matrix Differentiation

7 Residual Diagnostics

7.1 plot of residuals vs fitted values

Don't shake them up because then you may introduce patterns that don't reallt exist, the fitted values x axis is just for indexing position. Examine this by eye to make sure nothing is obviously wrong. Don't look at it too long. This is not enough to determine randomness, or normality of residuals. you will need tests and normal !! plots in addition to this.

7.2 ACF

Auto correlation plot: Will be good for forecasting. Blue bands measure serial correlation. for determining if the residuals are white noise you'd want about 95% of the spikes to be within the blue lines